

$K = $

$K_0 = \{\{a\}, \{b\}, \{c\}\}$

$K_1 = \{\{c,b\}\}$

$K_i = \emptyset \quad \forall\, i \geq 2$

↳ the set of i-cells

Chain Groups

coef group: $(G, +)$

$C_i = \langle K_i \rangle_{\text{pick an orientation for simplices}}$

$C_0 = \langle [a], [b], [c] \rangle = \langle [a], [a]+[b], [c] \rangle$ ← generating sets are not unique!

$= \{ \alpha[a] + \beta[b] + \gamma[c] \mid \alpha, \beta, \gamma \in G \}$

↑ set notation

with $G = \mathbb{Z}_2$

$C_0 \cong \mathbb{Z}^{K_0}$ (coef 1 = in the set, 0 = not in it)

with $G = \mathbb{Z}$

$C_0 = \{[a], [b], [c], [a]+[b], [a]+2[b], [a]-[b], 1[a]-3[b]+5[c], \ldots\}$

This is a group! We can add 2 cells.

$\alpha_1[a] + \beta_1[b] + \gamma_1[c] \in C_0$

$+ \alpha_2[a] + \beta_2[b] + \gamma_2[c] \in C_0$

$(\alpha_1+\alpha_2)[a] + (\beta_1+\beta_2)[b] + (\gamma_1+\gamma_2)[c]$

$C_1 = \langle [c,b] \rangle = \langle [b,c] \rangle$

$= \{\alpha [c,b] \mid \alpha \in G\}$

$\cong G$

with $\mathbb{Z}$ coef:

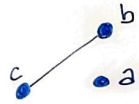
$\langle [c,b] \rangle = \{\alpha [c,b] \mid \alpha \in \mathbb{Z}\}$

$= \{\ldots, -2\cdot[c,b], -1\cdot[c,b], 0\cdot[c,b], 1\cdot[c,b], 2\cdot[c,b], \ldots\}$

$= \{\ldots, 2\cdot[b,c], 1\cdot[b,c], 0\cdot[b,c], -1[b,c], -2[b,c], \ldots\}$

$= \langle [b,c] \rangle$

$-1[c,b] = [b,c]$ ✱

$K =$ 

sidenote:

G, H groups

$f: (G,+) \longrightarrow (H,*)$

is called a (group) homomorphism if

$\forall\, g_1, g_2 \in G$

$f(g_1+g_2) = f(g_1) * f(g_2)$

$\Rightarrow$ Just need to define f on a generating set for G.

Boundary Operator

$\partial_0 : C_0 \to \{\emptyset$ (the one w/ the identity element)

$c \longmapsto \emptyset$

what is

$Z_0 := \ker(\partial_0) = \{c \in C_0 \mid \partial_0(c) = \emptyset\}$

$= C_0$

(Z_0: the cycle group ... things w/ no boundary)

$B_{-1} := \text{im}(\partial_0) = \text{range}(\partial_0)$

(B_{-1}: the boundary group... things that exist as boundaries of higher-d things)

$= \{\alpha \in C_{-1} \mid \exists c \in C_0 \text{ s.t. } \partial(c) = \alpha\}$

$= C_{-1}$

$\partial_1 : C_1 \longrightarrow C_0$

$[c,b] \longmapsto b - c$

note: Let $n \in \mathbb{Z}$

$n[c,b] \in C_1$

$\partial_1(n[c,b]) = n \cdot \partial([c,b])$

$= nb - nc$

$[c,b]$ generates C_1

$\therefore \forall\, x \in C_1, \exists n \in \mathbb{Z}$

$x = n \cdot [c,b]$

$Z_1 = \ker(\partial_1) = \{\emptyset\}$

$B_0 = \text{im}(\partial_1) = \{n[b] - n[c] \mid n \in \mathbb{Z}\} \cong \mathbb{Z}$

$\mathbb{Z}$ s.t.

general

$$\partial_i([v_0, v_1, v_2, \dots, v_i]) = \sum (-1)^j [v_0, v_1, \dots, \hat{v_j}, \dots, v_i]$$

$[v_0, v_1, v_2, \dots, v_i]$: i-dim simplex defined by i+1 vertices

$\hat{v_j}$: remove v_j

$Z_i = \ker(\partial_i)$ } These are groups!!

$B_{(i-1)} = \text{im}(\partial_i)$

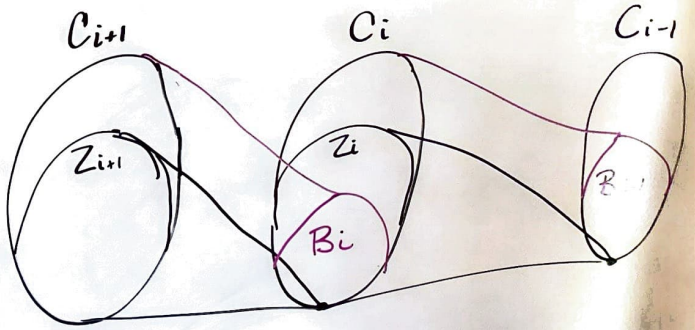
Z_i is a subgroup of C_i (use ∂_i to define)

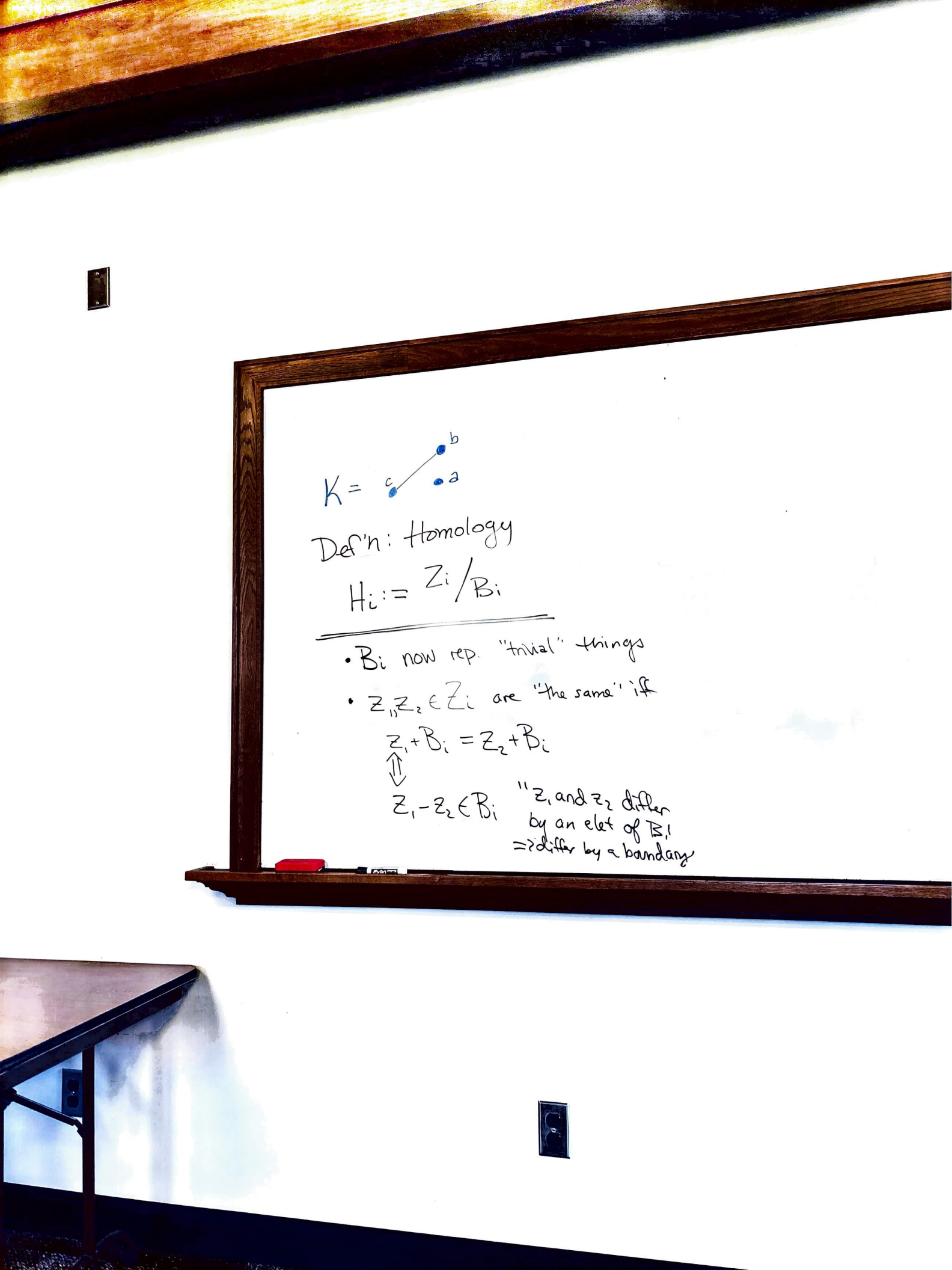
B_i is a subgroup of C_i (use ∂_{i+1} to define)

Q1: What is the relationship btwn Z_i and B_i?

A: $B_i \triangleleft Z_i$ Proof?

Q2: What happens when you take the boundary twice?




K =
b
c
a
Def'n: Homology
$H_i := Z_i / B_i$
• B_i now rep. "trivial" things
• $z_1, z_2 \in Z_i$ are "the same" iff
$z_1 + B_i = z_2 + B_i$
$\Updownarrow$
$z_1 - z_2 \in B_i$
"z_1 and z_2 differ by an elet of B_i ⇒ differ by a boundary